CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · PYQ SOLUTIONS

Chapter 9: Matrices

Step-by-step solutions to the previous year questions

The question numbers match the PYQ question sheet.

1 Multiplication of matrices

1. If $A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$, find A^2. (a) $\begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$ (b) $\begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix}$ (c) $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ (d) $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ ICSE 2025, 1 mark

SOLUTION

$$A^2 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 0+1 & 0+0 \\ 0+0 & 1+0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Answer: (c)

2. Given $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$, $B = \begin{bmatrix} 2 & 1 \\ 4 & 2 \end{bmatrix}$ and $C = \begin{bmatrix} -5 & 1 \\ 7 & -4 \end{bmatrix}$, find (a) $A + C$ (b) $B(A + C)$ (c) $5B$ (d) $B(A + C) - 5B$. ICSE 2025, 4 marks

SOLUTION

(a) $A + C = \begin{bmatrix} 1-5 & 2+1 \\ 3+7 & 4-4 \end{bmatrix} = \begin{bmatrix} -4 & 3 \\ 10 & 0 \end{bmatrix}$

(b) $B(A + C) = \begin{bmatrix} 2 & 1 \\ 4 & 2 \end{bmatrix}\begin{bmatrix} -4 & 3 \\ 10 & 0 \end{bmatrix} = \begin{bmatrix} -8+10 & 6+0 \\ -16+20 & 12+0 \end{bmatrix} = \begin{bmatrix} 2 & 6 \\ 4 & 12 \end{bmatrix}$

(c) $5B = \begin{bmatrix} 10 & 5 \\ 20 & 10 \end{bmatrix}$

(d) $B(A + C) - 5B = \begin{bmatrix} 2-10 & 6-5 \\ 4-20 & 12-10 \end{bmatrix}$

Answer: (d) $\begin{bmatrix} -8 & 1 \\ -16 & 2 \end{bmatrix}$

3. If $A = \begin{bmatrix} 2 & 2 \\ 0 & 2 \end{bmatrix}$ and $A^2 = \begin{bmatrix} 4 & x \\ 0 & 4 \end{bmatrix}$, find x. (a) 2 (b) 4 (c) 8 (d) 10 ICSE 2024, 1 mark

SOLUTION

$$A^2 = \begin{bmatrix} 2 & 2 \\ 0 & 2 \end{bmatrix}\begin{bmatrix} 2 & 2 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} 4+0 & 4+4 \\ 0+0 & 0+4 \end{bmatrix} = \begin{bmatrix} 4 & 8 \\ 0 & 4 \end{bmatrix}$$

Answer: (c)

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4. Given $A = \begin{bmatrix}1 & 3\\2 & 4\end{bmatrix}$, $B = \begin{bmatrix}1 & 2\\2 & 4\end{bmatrix}$, $C = \begin{bmatrix}4 & 1\\1 & 5\end{bmatrix}$ and $I = \begin{bmatrix}1 & 0\\0 & 1\end{bmatrix}$, find $A(B+C) - 14I$. ICSE 2023, 3 marks

SOLUTION

$$B + C = \begin{bmatrix}5 & 3\\3 & 9\end{bmatrix}$$

$$A(B+C) = \begin{bmatrix}5+9 & 3+27\\10+12 & 6+36\end{bmatrix} = \begin{bmatrix}14 & 30\\22 & 42\end{bmatrix}$$

$$14I = \begin{bmatrix}14 & 0\\0 & 14\end{bmatrix}$$

Answer: $\begin{bmatrix}0 & 30\\22 & 28\end{bmatrix}$

5. If $A = \begin{bmatrix}3 & 0\\5 & 1\end{bmatrix}$ and $B = \begin{bmatrix}-4 & 2\\1 & 0\end{bmatrix}$, find $A^2 - 2AB + B^2$. ICSE 2020, 3 marks

SOLUTION

$$A^2 = \begin{bmatrix}9+0 & 0+0\\15+5 & 0+1\end{bmatrix} = \begin{bmatrix}9 & 0\\20 & 1\end{bmatrix}$$

$$AB = \begin{bmatrix}-12+0 & 6+0\\-20+1 & 10+0\end{bmatrix} = \begin{bmatrix}-12 & 6\\-19 & 10\end{bmatrix}, \text{ so } 2AB = \begin{bmatrix}-24 & 12\\-38 & 20\end{bmatrix}$$

$$B^2 = \begin{bmatrix}16+2 & -8+0\\-4+0 & 2+0\end{bmatrix} = \begin{bmatrix}18 & -8\\-4 & 2\end{bmatrix}$$

$$A^2 - 2AB + B^2 = \begin{bmatrix}9+24+18 & 0-12-8\\20+38-4 & 1-20+2\end{bmatrix}$$

Answer: $\begin{bmatrix}51 & -20\\54 & -17\end{bmatrix}$

6. If $A = \begin{bmatrix}2 & 3\\5 & 7\end{bmatrix}$, $B = \begin{bmatrix}0 & 4\\-1 & 7\end{bmatrix}$ and $C = \begin{bmatrix}1 & 0\\-1 & 4\end{bmatrix}$, find $AC + B^2 - 10C$. ICSE 2018, 3 marks

SOLUTION

$$AC = \begin{bmatrix}2-3 & 0+12\\5-7 & 0+28\end{bmatrix} = \begin{bmatrix}-1 & 12\\-2 & 28\end{bmatrix}$$

$$B^2 = \begin{bmatrix}0-4 & 0+28\\0-7 & -4+49\end{bmatrix} = \begin{bmatrix}-4 & 28\\-7 & 45\end{bmatrix} \text{ and } 10C = \begin{bmatrix}10 & 0\\-10 & 40\end{bmatrix}$$

$$AC + B^2 - 10C = \begin{bmatrix}-1-4-10 & 12+28-0\\-2-7+10 & 28+45-40\end{bmatrix}$$

Answer: $\begin{bmatrix}-15 & 40\\1 & 33\end{bmatrix}$

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7. $A = \begin{bmatrix} 1 & 3 \\ 3 & 4 \end{bmatrix}$, $B = \begin{bmatrix} -2 & 1 \\ -3 & 2 \end{bmatrix}$ and $A^2 - 5B^2 = 5C$. Find the matrix C. ICSE 2017, 4 marks

SOLUTION

$$A^2 = \begin{bmatrix} 1+9 & 3+12 \\ 3+12 & 9+16 \end{bmatrix} = \begin{bmatrix} 10 & 15 \\ 15 & 25 \end{bmatrix}$$

$$B^2 = \begin{bmatrix} 4-3 & -2+2 \\ 6-6 & -3+4 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$5C = A^2 - 5B^2 = \begin{bmatrix} 5 & 15 \\ 15 & 20 \end{bmatrix}$$

Answer: $C = \begin{bmatrix} 1 & 3 \\ 3 & 4 \end{bmatrix}$

2 Solving matrix equations

8. $A = \begin{bmatrix} x & 0 \\ 1 & 1 \end{bmatrix}$, $B = \begin{bmatrix} 4 & 0 \\ y & 1 \end{bmatrix}$ and $C = \begin{bmatrix} 4 & 0 \\ x & 1 \end{bmatrix}$. Find x and y if $AB = C$. ICSE 2024, 4 marks

SOLUTION

$$AB = \begin{bmatrix} 4x+0 & 0+0 \\ 4+y & 0+1 \end{bmatrix} = \begin{bmatrix} 4x & 0 \\ 4+y & 1 \end{bmatrix}$$

Compare with C: $4x = 4$, so $x = 1$. Then $4 + y = x = 1$, so $y = -3$.

Answer: $x = 1, y = -3$

9. If $\begin{bmatrix} 2 & 0 \\ 0 & 4 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 2 \\ -8 \end{bmatrix}$, find x and y. (a) $1, -2$ (b) $-2, 1$ (c) $1, 2$ (d) $-2, -1$ ICSE 2023, 1 mark

SOLUTION

Multiply out: $\begin{bmatrix} 2x \\ 4y \end{bmatrix} = \begin{bmatrix} 2 \\ -8 \end{bmatrix}$, so $x = 1$ and $y = -2$.

Answer: (a)

10. $A = \begin{bmatrix} x & 3 \\ y & 3 \end{bmatrix}$ and $A^2 = 3I$, where I is the identity matrix of order 2. Find x and y. ICSE 2020, 3 marks

SOLUTION

$$A^2 = \begin{bmatrix} x^2+3y & 3x+9 \\ xy+3y & 3y+9 \end{bmatrix} = \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix}$$

$3x + 9 = 0$ gives $x = -3$, and $3y + 9 = 3$ gives $y = -2$.

Check: $x^2 + 3y = 9 - 6 = 3$ and $xy + 3y = 6 - 6 = 0$.

Answer: $x = -3, y = -2$

11. Given $\begin{bmatrix} 4 & 2 \\ -1 & 1 \end{bmatrix} M = 6I$, where I is the unit matrix of order 2×2, (a) state the order of M (b) find M.

ICSE 2019, 4 marks

SOLUTION

(a) The product is a 2×2 matrix and the first matrix is 2×2, so M is 2×2.

(b) Let $M = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$. Then $\begin{bmatrix} 4a+2c & 4b+2d \\ -a+c & -b+d \end{bmatrix} = \begin{bmatrix} 6 & 0 \\ 0 & 6 \end{bmatrix}$.

$-a + c = 0$ gives $c = a$, and then $6a = 6$, so $a = 1$ and $c = 1$.

$-b + d = 6$ gives $d = b + 6$, and then $4b + 2b + 12 = 0$, so $b = -2$ and $d = 4$.

Answer: $M = \begin{bmatrix} 1 & -2 \\ 1 & 4 \end{bmatrix}$

12. Given $B = \begin{bmatrix} 1 & 1 \\ 8 & 3 \end{bmatrix}$, find $X = B^2 - 4B$. Then solve $X \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} 5 \\ 50 \end{bmatrix}$ for a and b. ICSE 2017, 4 marks

SOLUTION

$B^2 = \begin{bmatrix} 1+8 & 1+3 \\ 8+24 & 8+9 \end{bmatrix} = \begin{bmatrix} 9 & 4 \\ 32 & 17 \end{bmatrix}$ and $4B = \begin{bmatrix} 4 & 4 \\ 32 & 12 \end{bmatrix}$

$X = \begin{bmatrix} 5 & 0 \\ 0 & 5 \end{bmatrix}$

$X \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} 5a \\ 5b \end{bmatrix} = \begin{bmatrix} 5 \\ 50 \end{bmatrix}$

Answer: $X = \begin{bmatrix} 5 & 0 \\ 0 & 5 \end{bmatrix}$; $a = 1, b = 10$

3 Addition, subtraction and multiplication by a number

13. Simplify $\sin A \begin{bmatrix} \sin A & -\cos A \\ \cos A & \sin A \end{bmatrix} + \cos A \begin{bmatrix} \cos A & \sin A \\ -\sin A & \cos A \end{bmatrix}$. ICSE 2019, 3 marks

SOLUTION

Multiply each matrix by its number, then add the matching entries.

$= \begin{bmatrix} \sin^2 A + \cos^2 A & -\sin A \cos A + \sin A \cos A \\ \sin A \cos A - \sin A \cos A & \sin^2 A + \cos^2 A \end{bmatrix}$

Answer: $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

14. Find x and y if $2 \begin{bmatrix} x & 7 \\ 9 & y-5 \end{bmatrix} + \begin{bmatrix} 6 & -7 \\ 4 & 5 \end{bmatrix} = \begin{bmatrix} 10 & 7 \\ 22 & 15 \end{bmatrix}$. ICSE 2018, 3 marks

SOLUTION

Left side $= \begin{bmatrix} 2x+6 & 14-7 \\ 18+4 & 2y-10+5 \end{bmatrix} = \begin{bmatrix} 2x+6 & 7 \\ 22 & 2y-5 \end{bmatrix}$

$2x + 6 = 10$ gives $x = 2$, and $2y - 5 = 15$ gives $y = 10$.

Answer: $x = 2, y = 10$

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